



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI ATHAYAMMAN FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

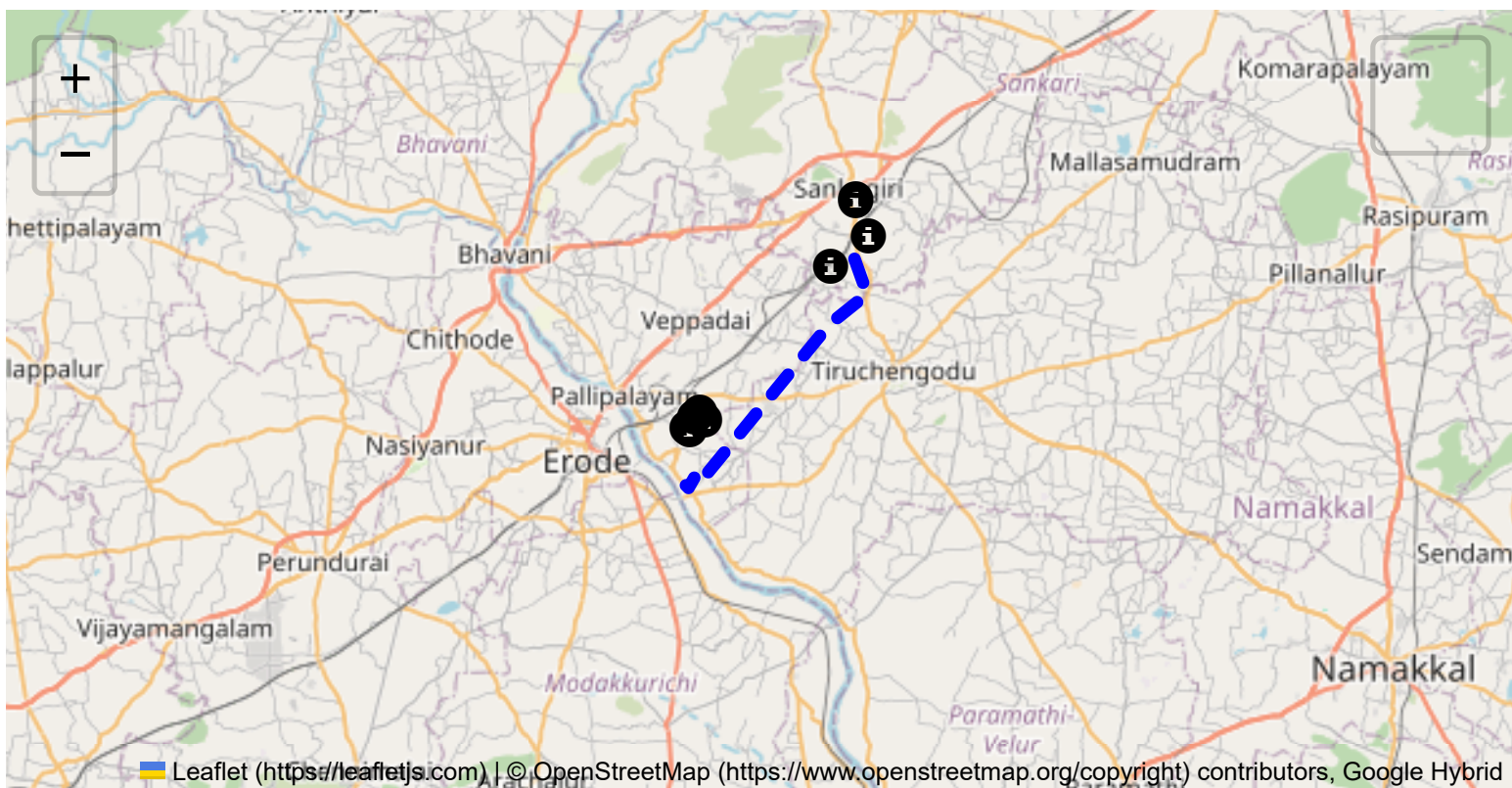
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

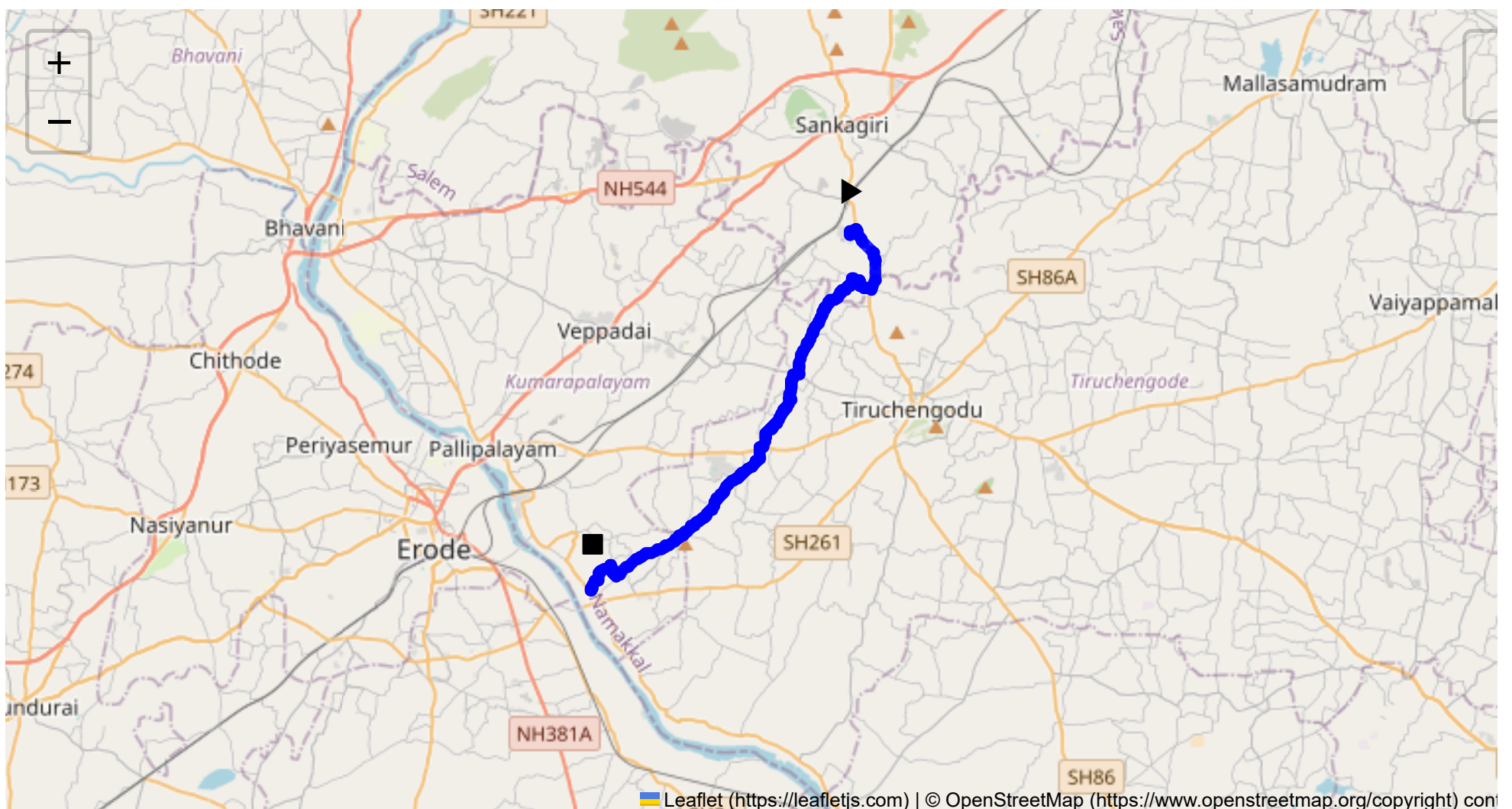
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 21.44 km
Estimated Duration: 0.7 hours
Adjusted Duration (Heavy Vehicle): 0.9 hours
Start: (11.4381, 77.8734)
End: (11.317287, 77.782477)



Welcome to the Journey Risk Management Study

- Overview of the Route Map:** The route is approximately 21.44 kilometers long, starting from Sangagiri and ending at Kokkarayanpettai Thevanampalayam. It passes through key areas such as Puthur, Palmadai, Karuveppampatti, Kokkarayanpet, and Devanampalayam. The journey is facilitated by regional roads and includes sections like State Highway 86 (SH 86).

2. **Typical Weather Conditions and Potential Weather-related Hazards:** Tamil Nadu experiences a tropical climate, with the potential for monsoon rains typically from October to December. Heavy rains can cause waterlogging, reduce visibility, and affect road conditions, particularly in low-lying areas.
3. **Analysis of Traffic Patterns:** Traffic congestion is typically observed during morning and evening peak hours, particularly around towns and market areas. SH 86 may experience higher traffic volumes due to its significance as a connector road.
4. **Assessment of Road Quality and Infrastructure:** The road quality varies, with some stretches well-maintained by state authorities like SH 86, while others might have potholes and narrower lanes, especially in village areas. Signage and lighting may be limited in rural stretches.
5. **Suggestions for Alternative Routes for Emergencies:** For emergencies, consider:
 - Diverting via larger highways like NH 544 if accessible.
 - Using local roads that parallel the main route, offering potential detours around congested or blocked sections.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Regulations require compliance with safety norms for transporting hazardous materials, including properly labeled and secured goods. Permission may be necessary for night transport in certain areas.
7. **Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:** While specific data may be limited, rural roads in the region have seen occasional accidents due to narrow lanes and sharp turns. Proper speed management and caution are advised.
8. **Environmental Considerations and Sensitive Areas:** Be mindful of agricultural zones and waterways, which could be affected by spills. Air and noise pollution near populated areas should be minimized, considering the proximity to villages and towns.
9. **Analysis of Communication Coverage:** Network coverage is generally reliable near towns but may become spotty in remote or rural stretches. It is advisable to have alternative communication means during the journey.
10. **Estimated Emergency Response Times:** Response times could vary; closer to towns, emergency services might reach within 30 minutes, while more rural or remote areas could take longer (1-2 hours). It's advisable to identify nearest emergency services en route.
11. **Overall Summary of Risk Assessment:** The route presents moderate risk mainly due to variable road conditions, potential congestion, and weather effects. Proper planning including updated weather reports, alternative routes for emergencies, and constant communication can help mitigate risks. Drivers should remain cautious about speed, especially in rural and populated areas, and comply with local regulations to ensure safety during transport.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41978, 77.88077	10 KM/Hr
5	Turn	High	11.41981, 77.88078	15 KM/Hr
6	Turn	Medium	11.42102, 77.87680	30 KM/Hr
7	Turn	Medium	11.42231, 77.87666	30 KM/Hr
8	Blind Spot	Blind Spot	11.42333, 77.87394	10 KM/Hr
9	Turn	Medium	11.42140, 77.87364	30 KM/Hr
10	Turn	Medium	11.41916, 77.87116	30 KM/Hr
11	Turn	High	11.41917, 77.87022	15 KM/Hr
12	Turn	Medium	11.41641, 77.86797	30 KM/Hr
13	Turn	High	11.40478, 77.86020	15 KM/Hr
14	Turn	High	11.40471, 77.86004	15 KM/Hr
15	Turn	Medium	11.39361, 77.85502	30 KM/Hr
16	Turn	High	11.39323, 77.85539	15 KM/Hr
17	Blind Spot	Blind Spot	11.39047, 77.85509	10 KM/Hr
18	Turn	High	11.39076, 77.85379	15 KM/Hr
19	Turn	Medium	11.38963, 77.85303	30 KM/Hr
20	Turn	Medium	11.38629, 77.85258	30 KM/Hr
21	Turn	Medium	11.38591, 77.85279	30 KM/Hr
22	Turn	High	11.38248, 77.85174	15 KM/Hr
23	Turn	Medium	11.38217, 77.85223	30 KM/Hr
24	Turn	Medium	11.37017, 77.84376	30 KM/Hr
25	Turn	High	11.36568, 77.84272	15 KM/Hr
26	Turn	High	11.36551, 77.84183	15 KM/Hr
27	Turn	Medium	11.36505, 77.84189	30 KM/Hr
28	Turn	Medium	11.36210, 77.84146	30 KM/Hr
29	Turn	Medium	11.36189, 77.84128	30 KM/Hr
30	Turn	Medium	11.36159, 77.84124	30 KM/Hr
31	Turn	Medium	11.33668, 77.81596	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
32	Turn	Medium	11.33552, 77.81470	30 KM/Hr
33	Turn	High	11.33554, 77.81355	15 KM/Hr
34	Turn	Medium	11.33480, 77.81344	30 KM/Hr
35	Turn	High	11.32143, 77.79174	15 KM/Hr
36	Turn	Medium	11.32349, 77.79012	30 KM/Hr
37	Turn	Medium	11.32378, 77.78975	30 KM/Hr
38	Blind Spot	Blind Spot	11.32490, 77.78918	10 KM/Hr
39	Turn	Medium	11.32408, 77.78895	30 KM/Hr
40	Turn	Medium	11.32243, 77.78555	30 KM/Hr
41	Turn	Medium	11.32013, 77.78492	30 KM/Hr
42	Turn	Medium	11.31955, 77.78388	30 KM/Hr
43	Blind Spot	Blind Spot	11.31627, 77.78274	10 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	K. S. Rangasamy College of Technology	11.3587246, 77.8278764	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.41978, 77.88077



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.42333, 77.87394



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.41917, 77.87022



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41641, 77.86797



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40478, 77.86020



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.40471, 77.86004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39361, 77.85502



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39323, 77.85539



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.39047, 77.85509



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.39076, 77.85379



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38963, 77.85303



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38629, 77.85258



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38591, 77.85279



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38248, 77.85174



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38217, 77.85223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.37017, 77.84376



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36568, 77.84272



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.36551, 77.84183



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.36505, 77.84189



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33668, 77.81596



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33552, 77.81470



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.33554, 77.81355



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33480, 77.81344



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.32143, 77.79174



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32349, 77.79012



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.32378, 77.78975



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.32490, 77.78918



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.32408, 77.78895



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32243, 77.78555



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.32013, 77.78492



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.31955, 77.78388



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.31627, 77.78274
